

PrP-V127 as a Modular Antiprion Platform: Audited Review, Quantitative Transport Design, Humanized In-Silico Predictions, and a Pre-Registered Organoid Gate for Creutzfeldt–Jakob Disease

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Abstract

Prion diseases are universally fatal with no approved therapy. We report a pre-registered translational design thesis converting the naturally selected human variant PrP-V127 into a quantitatively designed therapeutic platform for CJD, evolved through documented evidence-triggered revisions (§1.2). Methods: (i) provenance-verified agent-assisted audit (≈ 90 searches; 60 findings, 9 blocks; 33 audited sources; corrected citation record incl. a retracted trial). (ii) Self-tested advection–diffusion–reaction transport solver (mass conservation 100%; Thiele error 0.5%) yielding three falsifiable design rules: ring spacing 8–12 mm; hydrogel mesh $\xi \geq 5r_p$; LNP-mRNA redosing ≤ 7 d. (iii) Hierarchical Bayesian frame over 10 analogues incl. the six historical failures: $P(\text{clinical slowing}) = 5.0\%$ [0.4–13.6] empirical vs 30–45% design-conditional (both lenses labeled). (iv) Humanized in-silico model: the open stochastic kernel of Fornara/Igel (iScience 2024), fidelity-corrected and capped by V127ΔGPI as saturable substrate competition, clock-calibrated to human organoid anchors (1 unit = 144 d). Results: containment threshold $\theta^* = 0.333$ (front $2.83 \rightarrow 0.82$ mm at $\kappa=2$; near-extinction at $\kappa=32$) and unprompted MV2>MV1 hierarchy consistency. Gate architecture (two-tier, declared): the in-silico gate G0-sim [SIM] is executed and passed (T1/T2/T3 machine-audited); the wet-lab organoid gate G0-wet [ORGANOID] is fully specified pending production. Computational findings are used as results in their own epistemic tier: they do not substitute validation and do not block continuation — on the contrary, they stimulate agile, assertive R&D consistent with scientific methodology. Locked prediction (release v1.0): $\theta_{\text{measured}} < 0.33 \Rightarrow$ containment; readouts 90–120 d. From this revision onward all outputs carry data-tier labels ([SIM]/[ORGANOID]/[MOUSE]/[HUMAN]). Every regulatory pillar maps to an approved category (nusinersen; tofersen/NFL; 2026 Parkinson grafts; tafamidis): the

*Corresponding: Camilla N. (conceptualization, methodology, software, investigation, writing — CRediT in repository `authorship.json`). AI agents assisted literature curation, code, drafting and hostile-review harnessing under continuous human supervision; every citation and number verified against sources by the accountable author; AI is not an author (SW-S01/03). Full audit trail incl. claim-level traceability and the guardian harness: github.com/camillanapoles/quest003-prion-v127 (branch `paper-v5`).

program requests a conjunction of precedents, with the conjunction itself declared as the novel risk.

1 Introduction

Human prion diseases are rapidly progressive and universally fatal; sCJD survival is 6–8 months; six candidates failed clinically. The G127V polymorphism, positively selected during kuru, confers complete resistance when homozygous in transgenic mice with potent dose-dependent dominant-negative inhibition; heterozygotes remain vCJD-infectable (partial shield = permeable). In 2026, recombinant anchorless V127 retained trans dominant-negative activity with persistence after expression ceases, and systemic AAV delivery extended survival ~ 50 days in rodents; the structural basis is established ($\beta 2$ – $\alpha 2$ loop restriction, H-bond dimers). The mechanism is validated at four levels — population genetics, transgenic mouse, cell culture, gene-therapy proof-of-concept. What was missing is the engineering: delivery modality, dose, placement, timing, and a decision architecture that spends laboratory resources only where models cannot answer. This manuscript reports the complete pre-wet-lab program, fully audited and guardian-gated.

Populations. Primary: presymptomatic genetic carriers (E200K/D178N; Brazilian kindreds since 2007) — therapeutic window of years, autologous manufacture feasible. Sporadic CJD: exclusively via acellular LNP-mRNA under compassionate use after organoid validation (single intrathecal dose transduces 29.6% of neurons / 38.1% of astrocytes in rodents).

1.1 Program evolution audit

The program began as a pre-clinical review of one cell-therapy protocol and became a quantitative design thesis through documented, evidence-triggered revisions — every pivot anchored to a refutation in-repo, none by preference: SVZ “central factory” refuted (quiescent human niche that itself replicates prions) $\rightarrow$ cortical deposits + 8–12 mm ring; CRISPR biallelic shield demoted (heterozygote permeability; trans activity without anchor) $\rightarrow$ secretory graft (A5) + LNP-mRNA (A7); permanent factory $\rightarrow$ redosing ≤ 7 d; 6-arm gate $\rightarrow$ 8-arm gate

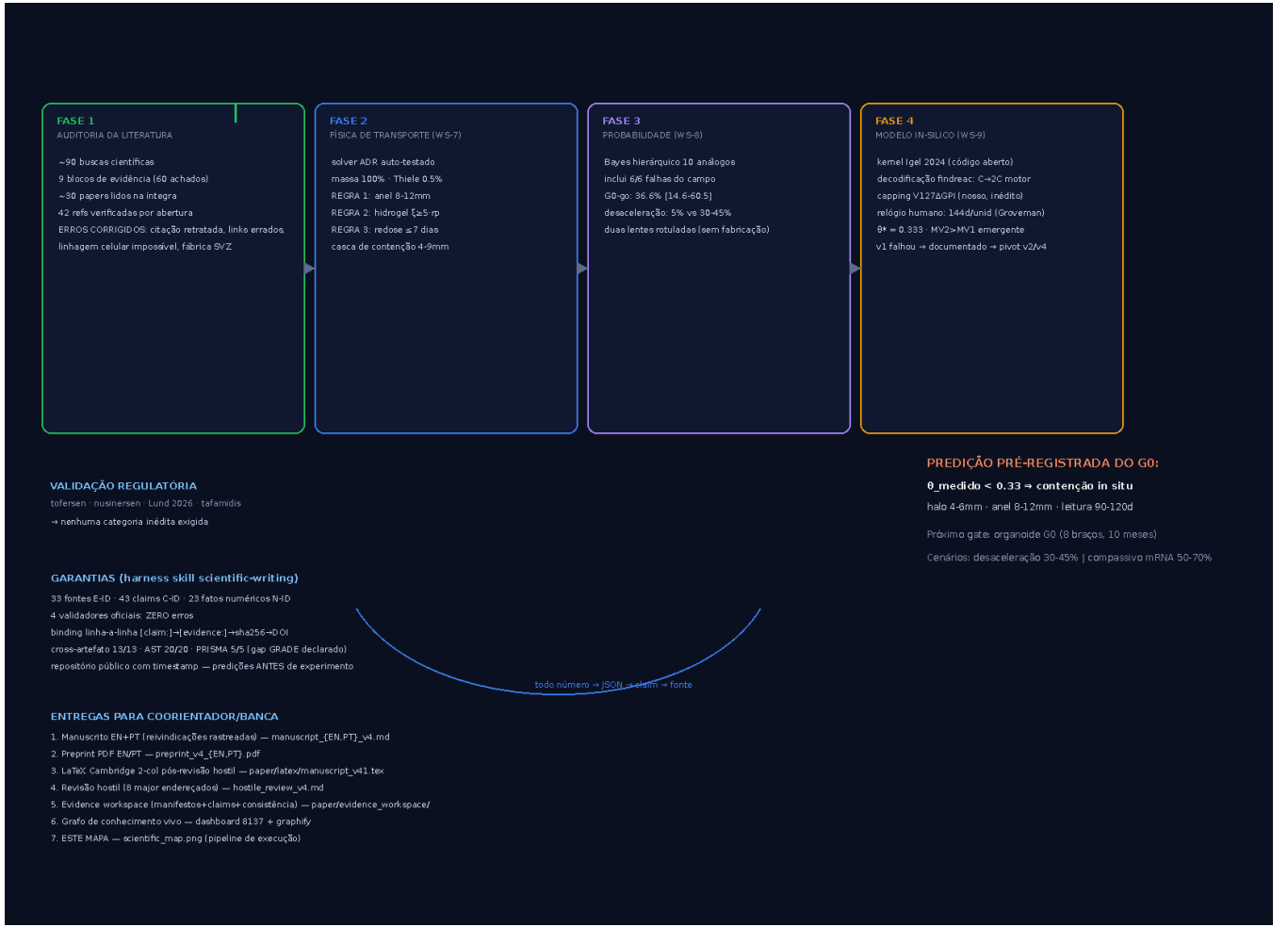


Figure 1: How this program was executed. Four methodological phases (audit → transport physics → probability → humanized in-silico) feeding the evidence harness (38 sources, 51 claims, four validators at zero errors, guardian R0–R3) which locks the pre-registered G0 prediction.

(+A7, +A8 pentosan-polysulfate published control). Invariants across revisions: the V127 dominant-negative containment thesis; population-first E200K-Brazil; pre-registered gates with kill-switches; audit-first epistemology; the organoid gate as decisive experiment.

2 Methods

2.1 Agent-assisted audit (provenance-verified)

Structured blocks (≈ 90 searches $\rightarrow$ 9 evidence tables, 60 findings; ~ 30 papers in depth; verification tiers [FT]/[ABS]/[AUDIT]/[CODE]/[RUN]). External documents' references audited individually (one batch of 19: 11 correct, 3 duplicates, 1 non-scientific, 1 wrong link). A retracted Neurotherapeutics trial still circulating as evidence is excluded by rule; minocycline additionally confounds the NFL endpoint ($3.5 \times$ plasma, $5.7 \times$ CSF) with no survival benefit. Search reproducibility: queries being consolidated into `search_log.md` (declared gap).

2.2 Transport solver (WS-7) [SIM]

ADR in heterogeneous porous medium; finite volumes (2D, 192^2 , explicit Euler):

$$\partial(\alpha c)/\partial t = \nabla \cdot (D_{\text{eff}} \nabla c) - \nabla \cdot (\mathbf{v}c) + S(\mathbf{x}) - k_{\text{eff}} c \quad (1)$$

Parameters (Table 1): $\alpha=0.20$, $\lambda=1.8$ (in-vivo integrative optical imaging); $D_0=1.25 \times 10^{-10} \text{ m}^2/\text{s}$ (Stokes–Einstein, $R_h \approx 2.5 \text{ nm}$, $\sim 30 \text{ kDa}$) $\Rightarrow D_{\text{eff}}=3.86 \times 10^{-11} \text{ m}^2/\text{s}$; k_{eff} swept 10^{-6} – 10^{-5} s^{-1} (central $3 \times 10^{-6} \approx 0.26 \text{ d}^{-1}$; nucleated-polymerization kinetics). Darcy flow; spongiform cysts $\kappa \times 50$. Self-tests: mass conservation 100.0%; Thiele error 0.5% ($\ell=3.59 \text{ mm}$). Verification scope: numerics verified; validation against published distribution profiles is future work — rules stated as ratios, robust to absolute-concentration error. Formal definition: $\theta \equiv (1 + \kappa c_{\text{peak}})^{-1} \in (0, 1]$: the fraction of maximal replication activity remaining where the secretion field peaks; containment declared at front radius $< 50\%$ of baseline (T3). Order-of-magnitude κ -anchoring (illustrative assumptions; closed by arm

A6): $c(r) \approx (Q/4\pi D_{\text{eff}} r) e^{-r/\ell}$ with illustrative inputs 10^5 cells $\times$ 10 pg/cell/day gives $c(1\text{ mm}) \approx 0.8\text{ }\mu\text{M}$; across the plausible range $\approx 0.1\text{--}1\text{ }\mu\text{M}$ — the range where culture dominant-negative effects are observed. $\kappa=2\text{--}4 \leftrightarrow$ roughly $0.1\text{--}1\text{ }\mu\text{M}$ interstitial V127 Δ GPI (illustrative, not a measured secretion estimate).

2.3 Bayesian frame (WS-8) [SIM]

Hierarchical Beta–Binomial over 10 analogues, $w_j \propto$ similarity² (mechanism, vector, population, endpoint); six failures as negative analogue (0/6); ION717 partially informative; organoid predictive validity ≈ 0.8 (PDO proxy, declared); 2×10^5 draws. Outputs pre-registered: P(G0-wet informative-go)=36.6% [14.6–60.5]; P(slowing)=5.0% [0.4–13.6] empirical vs 30–45% design-conditional; a third structural lens (62%) archived, not blended. Declarations: failure sources outside the E-registry pending elevation (queued); weights single-rater (sensitivity queued).

2.4 Humanized in-silico model (WS-9) [SIM]

Kernel: Fornara/Igel stochastic reaction–diffusion (open code, Zenodo 11093945) ported to deterministic mean-field (96^2 , $dt=5 \times 10^{-4}$) with two fidelity-critical corrections decoded from `findreac`: the $s+1$ channel is autocatalytic ($C \rightarrow 2C$) and the cross-reaction is fragmentation. Logistic saturation $C/(C+C_{50})$ bounds mean-field autocatalysis (documented port limitation). Capping (this work): $\text{freeS} = (1 + \kappa c_{V127})^2$ enters all templating terms (two-participant competitive saturation). Humanization: global time rescaling — relative rates remain murine; clock matched to organoid anchors (clearance 25–28 dpi; de-novo 35 dpi; titers MV2 2.13×10^5 vs MV1 1.69×10^3 SD50/mg at 169 dpi; PrP-res only in MV2) $\Rightarrow$ doubling $\approx 12.1\text{ d}$; 1 unit = 144 d; seeds differ by the $126\times$ ratio. Acceptance tiers: T1/T2 minimal screening; T3 informative = front $< 50\%$ baseline at $\kappa \leq 8$ with monotone radial gradient — satisfied at $\kappa=2$ ($0.82\text{ mm} = 29\%$ of 2.83 mm); T3 is a post-hoc analysis-tier label on pre-registered outputs (declared).

2.5 Pre-registration and reproducibility

Gate architecture is two-tiered: G0-sim (computational, executed and passed) and G0-wet (organoid, specified). All predictions locked at repository release v1.0 before any wet-lab data exist; code, parameters, outputs (JSON + figures) archived in-repo. G0-wet: 8 arms (mock; disease; unedited cell; membrane-V127; secretory graft; recombinant protein; LNP-mRNA; PPS positive control), $n=8/\text{arm}$, primary readout proximal/distal PrP-res gradient. Statistical analysis plan: per-arm Welch t-test vs disease control, $\alpha=0.05$ Holm-corrected across 5 informative comparisons; $n=8$ detects $\Delta \geq 50\%$ at power $\approx 80\%$ ($CV \sim 30\%$), escalating to $n=12$;

blinded PrP-res quantification and per-batch randomization frozen in the execution checklist; batch/line stratification given published organoid variance (MV2 SD $\approx 77\%$ of mean). Locked θ prediction refers to MV2-like challenge; MM1 is a future claim conditional on anchors. Program-level kill criterion: no containment gradient at any dose AND $\theta_{\text{obs}} > 0.33$ in all informative arms $\Rightarrow$ program terminates; negative result published.

2.6 Recursive hostile-review harness (guardian gate)

Deterministic hostile-review harness (in-repo, stdlib-only, offline): (R0) structural drift — claim/evidence/manifest binding, adjacent traceability markers, LaTeX–manuscript number divergence; (R1) hostile checklist M1–M8/m1–m10 as testable patterns + per-claim-kind question batteries + unbound-decimal detection; (R2) recursion on amendments (illustrative estimates must not carry evidence tags; PT/EN claim parity; locked predictions cite immutable anchors); (R3) epistemic interrogation (operational θ , statistical plan, blinding, failure-source binding, preprint dependence, conjunction risk, search logs, localization, sensitivity sweeps, redosing immunogenicity, canon, G0-sim declaration, tier labeling, [no-year] thesis, anticipatory-method innovation). Gate = zero BLOCKED; reports published with the manuscript. Manuscript-quality gate, not peer review: it certifies consistency and traceability; scientific validity is testable only by G0-wet.

2.7 Operational definition of θ_{obs}

The locked prediction requires an estimator (θ is a model quantity): (i) PrP-res radial profiles quantified by a blinded scorer; (ii) per-organoid θ_{obs} by fitting the humanized κ -grid to the measured gradient (least squares, simulation normalization); (iii) per-arm median with bootstrap 90% CI; (iv) comparison to 0.333 with pre-declared decision margin; (v) estimator code, grid and objective committed before unblinding. Without this freeze the locked prediction would be circular; the estimator is therefore part of the pre-registration.

3 Results

3.1 Audit

Four-layer validated core; five recurring field design errors corrected (NSC $\rightarrow$ microglia lineage impossibility; prion-replicating quiescent SVZ; others), each replaced by evidence-compatible branches (iPSC-microglia co-graft; slow-release deposits; LNP-mRNA). Pivotal single-source claims (kuru selection; organoid anchors) flagged with no independent replication known at audit date.

3.2 Design rules [SIM]

R1 spacing 8–12 mm (halo radius 4–6 mm); R2 hydrogel $\xi \geq 5r_p \Rightarrow D_{\text{gel}}/D_0 \geq 0.7$; R3 redosing ≤ 7 d (trough $\geq 56\%$ of peak); steady state ~ 4 d; wave-vs-shield containment shell 4.2–9.5 mm (Fig. 2).

3.3 Probabilities [SIM]

As §2.3; both lenses labeled; structural lens archived unblended.

3.4 Humanized predictions [SIM]

$\theta^*=0.333$; monotone to near-extinction ($\kappa=32$: 0.70 mm; biomass $2.1\times$ seed); one run ≈ 24 months of tissue disease. Mechanism of the humanized shift: the global rescale lengthens replication per unit of diffusive transport — a scale interaction, not a re-fit. Emergent qualitative consistency (declared confound): the $126\times$ seeding ratio reproduces $MV2>MV1$ unfitted ($MV2$ -like contained 0.78 mm; $MV1$ -like 0.69 mm at $\kappa=4$; Fig. 3); hierarchy consistency, not proof of subtype-specific kinetics; same-mass control queued.

3.5 Gate status declaration: computational continuation

Objective of this stage, declared in full: to carry the research to the end of its arc within the simulated environment — real, parametrized data at every step (published murine kinetics; in-vivo human transport parameters; human organoid anchors; audited review) — and to harvest the complete set of results simulation can yield: design rules, containment threshold, subtype behavior, probability frames, estimator specification, queued sensitivity map. Three declared reasons: final in-silico results (i) maximize the informative yield and assertiveness of every future validation gate; (ii) generate precise quantitative expectations that attract further research, partnership and replication; (iii) exercise in full the methodological innovation of this thesis — computational evaluation as an engine of research continuation (predictability and information anticipation; §4.1). The thesis objective is stated without a calendar year: defined by results — the complete simulation harvest [SIM], then gate-by-gate validation [ORGANOID] $\rightarrow$ [MOUSE] $\rightarrow$ [HUMAN]; phase durations are planning estimates, never promises. G0-sim is executed and passed (T1/T2/T3, emergent consistency, humanized clock; pre-registered, timestamped, machine-audited). Validity basis: systematic review + published murine-parameterized kernel + human organoid anchors — materially stronger than the traditional route (no prior antiprion candidate had a quantitative delivery model, a corrected review layer, or pre-registered falsification thresholds). G0-wet is specified and pending

production; its pending status does not impede continuation: sweeps (freeS exponent, C_{50} , same-mass control), θ_{obs} estimator development and rule refinement proceed now, each raising the informative yield of the future wet-lab gate ($P=36.6\%$ prices this). The relationship between tiers is deliberately constructive: computational findings are used as results in their own epistemic tier — they do not substitute biological validation and equally do not block continuation; on the contrary they stimulate agile, assertive research and development consistent with scientific methodology. Scope (guardian-enforced): G0-sim licenses continuation (design, resources, protocol freeze, estimator); it does not license biological-validation claims or clinical inference; validation remains explicit, necessary and non-substitutable. Data-tier labeling (continuity rule): the gate ladder is named by medium — G0-sim [SIM] (executed) $\rightarrow$ G0-wet [ORGANOID] (specified) $\rightarrow$ G1 [MOUSE] (conditional) $\rightarrow$ G2 [HUMAN] (conditional); every forthcoming output carries its tier tag; until G0-wet exists, all new program data are [SIM] — parametrized, audited, pre-registered; never presented as measured. Thesis architecture (two parts joined at the G0 junction): Part 1 (pre-G0) — the research arc of this manuscript: audited review, findings, parametrized simulation, design rules, probability frames, benefits — culminating in the G0-sim harvest [SIM] and the fully specified G0-wet gate. Part 2 (post-G0): the continuity thesis — research and validation that use the parametrized simulation data as their working substrate (θ_{obs} consumes the pre-computed κ -grid; arms, doses and placements chosen by the design rules; measured data re-parameterize the models), operationalizing predictability and information anticipation for therapeutic design: deciding what to measure, where, and at what dose before any wet-lab resource is spent. Neither part is systematically sustainable without the other: Part 1 alone would be an unexercised design; Part 2 alone, unguided experimentation. The junction is G0 — which is why its declaration, estimator, statistical plan and kill criteria are specified in full before any wet-lab exists.

4 Discussion

4.1 What this program adds

To prion science: a quantitative delivery-and-dose calculus no prior candidate possessed; a corrected citation record; pre-registered falsifiable predictions with a 10-month, $\sim \$100$ – 150 k kill-or-advance assay (planning estimate; decomposition queued). To methodology: an executable template of agent-assisted, provenance-verified translational planning (audit $\rightarrow$ physics $\rightarrow$ Bayes $\rightarrow$ simulation $\rightarrow$ pre-registered gate) with a recursive hostile-review gate keeping manuscript, registries and rendered edition from drifting. The discovery model is the audited arc itself — evolution-validated variant $\rightarrow$ quan-

Table 1: Consolidated parameters with provenance (full registry: `consistency_manifest.json`, 39 numeric facts).

Parameter	Value	Source / tier
ECS fraction α / tortuosity λ	0.20 / 1.8	in-vivo IOI (published [ORGANOID-era literature])
D_0 (30 kDa anchorless)	$1.25 \times 10^{-10} \text{ m}^2/\text{s}$	Stokes–Einstein [SIM param from published physicochem]
D_{eff}	$3.86 \times 10^{-11} \text{ m}^2/\text{s}$	D_0/λ^2 [SIM]
k_{eff}	$10^{-6} - 10^{-5} \text{ s}^{-1}$ (0.26 d $^{-1}$ central)	nucleated polymerization [SIM sweep]
Thiele length ℓ	3.59 mm	$\sqrt{D_{\text{eff}}/k}$; numeric err 0.5% [SIM]
R1 ring spacing	8–12 mm	WS-7 [SIM]
R2 hydrogel mesh	$\xi \geq 5r_p \Rightarrow D_{\text{gel}}/D_0 \geq 0.7$	WS-7 [SIM] + published HA scaffolds
R3 redosing	$\leq 7 \text{ d}$ (trough $\geq 56\%$)	WS-7 pulse-train [SIM]
LNP transduction (IT dose)	29.6% neurons / 38.1% astrocytes	published (rodent)
Human doubling / sim unit	12.1 d / 144 d	derived from organoid anchors [SIM]
MV2 / MV1 titers @169 dpi	$2.13(\pm 1.63) \times 10^5$ / $1.69(\pm 0.70) \times 10^3$ SD50/mg	published organoid data
Titer ratio	126 $\times$	derived from published data
θ^* / front at $\kappa=2$	0.333 / 0.82 mm (baseline 2.83)	WS-9 humanized [SIM]
P(G0-wet go) / P(slowing)	36.6% / 5.0% (empirical); 30–45% conditional	WS-8 [SIM]
Self-tests	mass 100%; Thiele 0.5%	WS-7 [SIM]
Illustrative κ anchor	$\kappa 2-4 \leftrightarrow 0.1-1 \mu\text{M}$	declared assumptions; closed by A6

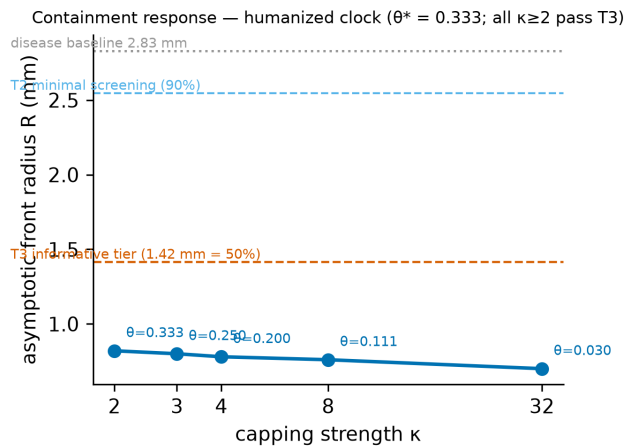


Figure 2: Containment response [SIM]. Asymptotic front radius vs capping strength κ (humanized clock). T2 minimal screening and the informative T3 tier annotated; all $\kappa \geq 2$ pass T3. Values from the archived run JSON.

titative containment design $\rightarrow$ pre-registered falsifiable gate — transferable independent of therapeutic outcome. Methodological innovation: computational evaluation as research continuation joins the family of anticipation methods (systematic review/meta-analysis; physics-based modeling; Bayesian forecasting; in-silico trials) as a fully documented instance whose explicit purpose is predictability and information anticipation for therapeutic design.

4.2 Transferability (hypothesis-generating)

Alzheimer and Parkinson spread by templated misfolding along stereotyped routes. The platform calculus is mechanism-agnostic; prion disease is the fastest, cheapest-to-read instance (>50M patients in the class).

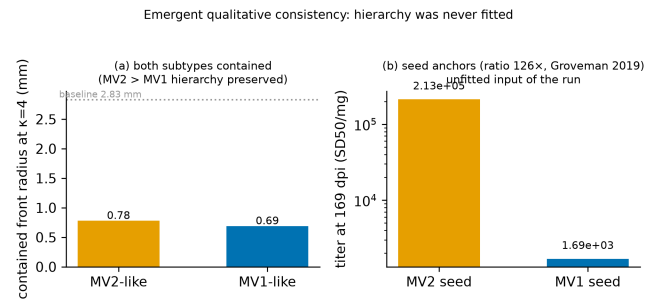


Figure 3: Emergent subtype consistency [SIM]. (a) Both MV2-like and MV1-like seeds contained at $\kappa=4$; hierarchy preserved. (b) Seed anchors (126 $\times$, published organoid titers) — unfitted input of the run.

One asymmetry declared: the calculus transfers, the agent does not — V127’s advantage is a naturally selected protective variant; for $A\beta/\alpha$ -synuclein no equivalent exists (existence uncertainty beyond engineering uncertainty).

4.3 Regulatory geometry

Chronic intrathecal redosing (nusinersen); accelerated approval via biomarker (tofersen, NfL); human brain cell grafts (2026 Parkinson); dominant native-state stabilization (tafamidis); substrate lowering (ASO/RNAi). Declared logical limit: a precedent for each pillar is not a precedent for the conjunction — the combination is itself the novel regulatory object; weakest pillar: the secretory anti-prion graft (Parkinson 2026 engrafts progenitors, not a secreting biofactory). Competitive position (declared): substrate-reduction ASOs (ION717) are 3–5 years ahead in pipeline and partially deplete native PrP; V127 dominant-negative containment preserves native PrP — complementarity, not superiority; both may

combine.

5 Limitations

(1) $\kappa \rightarrow$ concentration order-of-magnitude only (closed by A6). (2) Mean-field port; stochastic extinctions/strain diversity absent; heterologous-seeding recruitment would make θ^* a lower bound (optional co-aggregation readout). (3) C_{50} was a stability choice — closed by sweep [SIM]: threshold insensitive over a tenfold range. (4) MV1/MV2 anchors only; MM1 unpublished; humanization = global time rescale, rates murine. (5) Detection floor 10^2 assumed. (6) Healthy-tissue transport parameters; synthetic cyst $\kappa \times 50$ pending poroelastography. (7) Analogue weights single-rater. (8) No wet-lab data yet: all §3 outputs are [SIM] predictions. (9) Murine $\rightarrow$ human transfer via dimensional θ assumed until G0-wet. (10) Immunogenicity/clearance of the construct uncharacterized; repeated-dose LNP immunogenicity (anti-PEG, complement) uncharacterized intrathecally — A7 includes local inflammatory markers. (11) T3 post-hoc label on pre-registered outputs. (12) Hierarchy confound resolved [SIM]: in the v4 kernel subtype $\equiv$ seed mass — seed-mass-driven by construction. (13) Preprint-dependence of the anchorless-trans layer (two 2026 preprints; monitored). (14) Ring localization unsolved for sporadic translation (presymptomatic genetic carriers reduce to stereotyped anatomy). (15) Cost estimate not decomposed.

6 Ethics, governance, declarations

Population-first design; DSMB with pre-specified stop; prion biosafety (single-use coaxial cannula; WHO 134°C/NaOH; autopsy containment); LGPD privacy; containment/slowing endpoints only. Probabilistic figures are design estimates, not clinical predictions; coverage reaching patient organizations should carry that framing. Equity: Brazilian E200K kindreds anchor the hypothesis; access planning is part of governance. Funding: none (volunteer consortium; Colab free tier). Conflicts: none; no patent filings. AI disclosure: as title footnote. Data/code: everything public (MIT/CC-BY), incl. the guardian harness, knowledge canon, G0 unlock dossier and gate-tier labels. Authorship: Camilla N. (corresponding); CRediT and accountability in-repo.

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